BSCDS45: CLOUD COMPUTING

BSc Data Science. IV Year VII Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Basic understanding of business and mathematics

Course Objective:
The aim of this course is basic understanding of cloud concepts and Linux command-line operations is helpful. These prerequisites ensure readiness to explore cloud computing topics like IaaS, PaaS, SaaS, cloud security, and hands-on activities with platforms like AWS, Azure, and Google Cloud.

Learning Course Outcome: At the end of the course the students will be able to
CO1: Understand the evolution, components, and characteristics of cloud computing.
CO2: Explore cloud infrastructure services, architecture, and deployment models.
CO3: Gain hands-on experience with cloud simulators like CloudSim and GreenCloud.
CO4: Analyze cloud security challenges, risks, and mitigation strategies.

Course Contents

UNIT-I	Introduction Evolution of Cloud Computing	9 Hours
Origins of Cloud computing – Cloud components - Essential characteristics – On-demand self-service, Broad network access, Location independent resource pooling, Rapid elasticity , Measured service, Comparing cloud providers with traditional IT service providers, Roots of cloud computing.		
UNIT-II	**Infrastructure Services**	**9 Hours**
Architectural influences – High-performance computing, Utility and Enterprise grid computing, Cloud scenarios – Benefits: scalability, simplicity, vendors, security, Limitations – Sensitive information - Application development- security level of third party - security benefits, Regularity issues: Government policies.		
UNIT-III	**Cloud Architecture**	**9 Hours**
Layers and Models Layers in cloud architecture, Software as a Service (SaaS), features of SaaS and benefits, Platform as a Service (PaaS), features of PaaS and benefits, Infrastructure as a Service (IaaS), features of IaaS and benefits, Service providers, challenges and risks in cloud adoption. Cloud deployment model: Public clouds – Private clouds – Community clouds - Hybrid clouds - Advantages of Cloud computing.		
UNIT-IV	**Cloud Simulators**	**9 Hours**
CloudSim and GreenCloud Introduction to Simulator, understanding CloudSim simulator, CloudSim Architecture (User code, CloudSim, GridSim, SimJava) Understanding Working platform for CloudSim, Introduction to GreenCloud		
UNIT-V	**Cloud Security Considerations**	**9 Hours**

STRIDE Threat Model - Cloud Security Challenges – Cloud specific Cryptographic Techniques – CIA Triad – Security by Design – Common Security Risks - Risk Management – Security Monitoring – Security Architecture Design – Data Security – Application Security – Virtual Machine Security.

TEXTBOOKS:

1. "Cloud Computing: Concepts, Technology & Architecture" by Thomas Erl et al.
2. "Cloud Computing: From Beginning to End" by Ray J. Rafaels.

REFERENCES:

1. "Cloud Security and Privacy" by Tim Mather et al.
2. "CloudSim: A Toolkit for Modeling and Simulation of Cloud Computing Environments" by RajkumarBuyya et al.
3. "AWS Certified Solutions Architect Study Guide" by Ben Piper and David Clinton